

THE SMALL RAMSEY DEGREE OF AN EDGE IN ORDERED SPLIT GRAPHS IS THREE

ZIJIAN ZENG

ABSTRACT. We determine the small Ramsey degree of the ordered edge in the class of finite linearly ordered split graphs with induced order-preserving embeddings: it is exactly three. For the upper bound, a chosen clique-independent-set partition turns a split graph into an ordered free-amalgamation structure. The three expansions of an ordered edge, of types QQ , QI , IQ , are homogenized successively and the partition is then forgotten. For sharpness, an ordered four-vertex path has a unique split partition and forces all three types in every host. We also prove that the corresponding degree in the full class of ordered chordal graphs is at least two, and exhibit exact failures of free amalgamation and of amalgamation after naming a perfect elimination ordering. These diagnostics explain why the same argument does not settle the chordal problem, which remains open.

1. INTRODUCTION

The KAMAK 2025 problem collection records the following question as Problem 14: what is the small Ramsey degree of an edge in the class of finite linearly ordered chordal graphs under induced embeddings [4, Problem 14]? The problem is listed as suggested by Maximilian Strohmeier and as proposed by Nešetřil in 2025. Its closest result in the collection is vertex indivisibility of chordal graphs, proved by Guingona, Nusbaum, Padamsee, Parnes, Pippin, and Zinman [2, Proposition 4.13]. Searches of exact phrases and the structural Ramsey literature through August 28, 2026, located no subsequent solution, so we treat the full edge problem as open.

This note gives an exact answer for the subclass of split graphs, one of the restricted classes proposed as a first test case. Recall that a graph is split if its vertices can be partitioned into a clique and an independent set. Our main result is the following.

Theorem 1.1. *Let A be the two-vertex ordered graph with one edge. In the class $\mathcal{S}_{<}$ of finite linearly ordered split graphs with induced order-preserving embeddings,*

$$t_{\mathcal{S}_{<}}(A) = 3.$$

The value does not follow merely by proving a Ramsey theorem for split graphs with a named partition: the partition is not part of the category in Theorem 1.1, and split partitions need not be unique. We therefore make the forgetful step explicit. The upper bound uses the classical partite-construction theorem for ordered relational structures of Nešetřil and Rödl [5], in the ordered free-amalgamation form of Hubička and Nešetřil [3, Corollary 4.2]. The lower bound is not a formal expansion count: it uses a target whose split partition is forced in every unmarked copy.

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For the original chordal class $\mathcal{C}_{<}$, we obtain the rigorous but nonmatching bound

$$tc_{<}(A) \geq 2.$$

We also give small amalgamation obstructions showing why a direct PEO expansion or an unmodified free-amalgamation partite construction cannot supply the missing upper bound. We make no claim that the full chordal degree is finite.

2. DEFINITIONS

All graphs are finite, simple, undirected graphs. A *linearly ordered graph* is a pair $(G, <)$, where $<$ is a linear order on $V(G)$. An embedding $f : (G, <_G) \rightarrow (H, <_H)$ is injective and satisfies

$$x <_G y \iff f(x) <_H f(y), \quad xy \in E(G) \iff f(x)f(y) \in E(H).$$

Thus all graph embeddings below are induced and order preserving.

For ordered structures A, B, C , positive integers r, d , and the relevant class understood, write

$$C \longrightarrow (B)_{r,d}^A$$

if every coloring $\chi : \text{Emb}(A, C) \rightarrow [r]$ admits $b \in \text{Emb}(B, C)$ for which

$$|\{\chi(b \circ a) : a \in \text{Emb}(A, B)\}| \leq d.$$

The *small Ramsey degree* $t_{\mathcal{K}}(A)$ is the least positive integer d , if it exists, such that for every $B \in \mathcal{K}$ and every $r \geq 1$, some $C \in \mathcal{K}$ satisfies the displayed arrow. If no such d exists, the degree is ∞ . Since the ordered edge A is rigid, its copies are canonically the edges of the host.

A *split partition* of a graph G is a decomposition $V(G) = Q \sqcup I$ in which Q induces a clique and I an independent set. A *chordal graph* has no induced cycle of length at least four. We use the convention that a *perfect elimination ordering* (PEO) $\prec$ has the property that the $\prec$ -earlier neighbors of every vertex form a clique. Chordal graphs are exactly the graphs admitting a PEO [1].

3. THE MARKED RAMSEY CLASS

Let $\mathcal{S}_{<}^*$ be the class of ordered split graphs equipped with a named split partition $Q \sqcup I$, with embeddings required to preserve the parts. We encode this class in the finite relational language

$$L = \{<, Q, R\}.$$

Here Q is a unary predicate, I is its complement, and $R(q, i)$ records an arbitrary cross edge from $q \in Q$ to $i \in I$. The only allowed R -tuples satisfy $R(x, y) \Rightarrow Q(x) \wedge \neg Q(y)$; apart from this typing condition, R is unrestricted. The graph is recovered by the quantifier-free formula

$$(1) \quad E(x, y) \iff (Q(x) \wedge Q(y)) \vee (Q(x) \wedge \neg Q(y) \wedge R(x, y)) \vee (Q(y) \wedge \neg Q(x) \wedge R(y, x)).$$

In particular, an L -embedding forgets to an induced graph embedding, not only to a homomorphism.

Proposition 3.1. *The class $\mathcal{S}_{<}^*$ is a Ramsey class.*

Proof. Before adding $<$, the encoded structures form a two-sorted free amalgamation class: take the disjoint union over the common substructure and add no new R -tuples. After adding an arbitrary linear order, free amalgams are obtained by completing the two inherited orders to any common linear order. (Using a strict order instead of its reflexive closure is a definitionally equivalent convention.) Thus the class has the ordered free amalgamation property, and the proposition follows from the ordered free-amalgamation Ramsey theorem [3, Corollary 4.2].

The same source explicitly notes that bipartite graphs with one side named by a unary relation have precisely this free-amalgamation representation [3, p. 52]. $\square$

In the original graph language, pairs of new Q -vertices in an amalgam must be joined, so calling that presentation literally free would be incorrect. Formula (1) removes the issue: clique edges are definable rather than freely chosen relation tuples.

The ordered edge has exactly three expansions in $\mathcal{S}_{<}^*$:

$$A_{QQ}, \quad A_{QI}, \quad A_{IQ}.$$

The subscripts give the parts of the smaller and larger endpoints, respectively. There is no A_{II} , because I is independent.

Lemma 3.2 (Simultaneous homogenization). *Fix $B^* \in \mathcal{S}_{<}^*$, let $T \subseteq \{QQ, QI, IQ\}$ be the edge types that occur in B^* , and let $r \geq 1$. There exists $C^* \in \mathcal{S}_{<}^*$ such that, for every simultaneous r -coloring of the copies of each A_τ , $\tau \in T$, there is a copy of B^* on which the coloring of each fixed type A_τ is constant.*

Proof. Write $T = \{\tau_1, \dots, \tau_m\}$. Put $D_0 = B^*$, and use Proposition 3.1 successively to choose

$$D_j \longrightarrow (D_{j-1})_{r,1}^{A_{\tau_j}} \quad (1 \leq j \leq m).$$

Set $C^* = D_m$. Apply these arrows in reverse order. First find a copy of D_{m-1} homogeneous for type τ_m , then work inside it to homogenize τ_{m-1} , and continue. Passing to a further subcopy preserves all homogeneity obtained earlier. $\square$

4. THE UPPER BOUND AFTER FORGETTING THE PARTITION

Proposition 4.1. *For every $B \in \mathcal{S}_{<}$ and $r \geq 1$, there is $C \in \mathcal{S}_{<}$ such that*

$$C \longrightarrow (B)_{r,3}^A.$$

Proof. Choose one split partition of B , producing $B^* \in \mathcal{S}_{<}^*$, and apply Lemma 3.2. Let C be the unmarked reduct of the resulting C^* . Given an arbitrary r -coloring of $E(C)$, color a copy of each A_τ by the color of its underlying edge. Every edge of C has exactly one of the types QQ, QI, IQ . The lemma yields a marked copy of B^* whose edge color depends only on its type, so it uses at most three colors.

By formula (1), forgetting the marks on this copy gives an induced order-preserving copy of B . Notice that no expansion property is being assumed: we fixed one expansion B^* and constructed a marked host that contains that expansion. The expansion is precompact in any case, since an n -vertex graph has at most 2^n split partitions. $\square$

5. THE SHARP LOWER BOUND

Let P be the path

$$(2) \quad i_1 - q_1 - q_2 - i_2, \quad i_1 < q_1 < q_2 < i_2.$$

Lemma 5.1. *The graph P has the unique split partition*

$$Q = \{q_1, q_2\}, \quad I = \{i_1, i_2\}.$$

Proof. In any split partition, Q is a vertex cover because I is independent. Since P has no triangle, $|Q| \leq 2$, while no single vertex covers its three edges. Of the two-vertex covers, only $\{q_1, q_2\}$ is a clique. The claimed partition follows. $\square$

Proposition 5.2. *For every ordered split graph C , there is a three-coloring of $E(C)$ for which every induced order-preserving copy of P uses all three colors. Consequently $t_{\mathcal{S}_{<}}(A) \geq 3$.*

Proof. Choose any split partition $V(C) = Q_C \sqcup I_C$. For an edge xy with $x < y$, assign the color

$$\chi(xy) = \begin{cases} QQ, & x, y \in Q_C, \\ QI, & x \in Q_C, y \in I_C, \\ IQ, & x \in I_C, y \in Q_C. \end{cases}$$

These cases exhaust the edges because I_C is independent.

Let $f : P \rightarrow C$ be an induced order-preserving embedding. Intersecting its image with Q_C and I_C , then pulling back, gives a split partition of P . Lemma 5.1 forces $f(q_1), f(q_2) \in Q_C$ and $f(i_1), f(i_2) \in I_C$. Order preservation and (2) give the three edge colors

$$\chi(f(i_1)f(q_1)) = IQ, \quad \chi(f(q_1)f(q_2)) = QQ, \quad \chi(f(q_2)f(i_2)) = QI.$$

Thus every copy uses all three colors. If C has no copy of P , the Ramsey arrow fails already. Hence for this fixed P and $r = 3$, no ordered split host satisfies $C \rightarrow (P)_{3,2}^A$. $\square$

Propositions 4.1 and 5.2 prove Theorem 1.1.

6. WHAT REMAINS FOR CHORDAL GRAPHS

Split graphs are chordal, but Theorem 1.1 is not a lower bound of three for the larger category: a chordal Ramsey host is not required to be split. The following universal coloring gives the presently certified lower bound for the full class.

Proposition 6.1. *For the class $\mathcal{C}_<$ of finite linearly ordered chordal graphs with induced order-preserving embeddings,*

$$t_{\mathcal{C}_<}(A) \geq 2.$$

Proof. For an arbitrary host $(C, <)$, choose a PEO $\prec$. Color an edge xy , $x < y$, by 0 if $x \prec y$, and by 1 otherwise. The restriction of a PEO to an induced subgraph is again a PEO.

Take the path with edges ab, bc, cd , displayed in the order

$$c < a < d < b.$$

This order is not a PEO: the last vertex b has earlier nonadjacent neighbors a, c . Its reverse is not a PEO either: then c has earlier nonadjacent neighbors b, d . If an induced ordered copy of the path used only color 0, its displayed order would be a PEO; if it used only color 1, the reverse displayed order would be a PEO. Therefore every copy uses both colors, proving the lower bound. $\square$

We next isolate two obstructions to standard upper-bound strategies.

Proposition 6.2 (Failure of free amalgamation). *The class of chordal graphs is not closed under free amalgamation of induced subgraphs.*

Proof. Let the common subgraph consist of two nonadjacent vertices u, v . In one extension add a vertex x adjacent to u, v , and in the other add y adjacent to u, v . Each extension is a three-vertex path. Their free amalgam contains exactly the induced cycle $u - x - v - y - u$. $\square$

Adding the chord xy repairs this small amalgam, but arbitrary chordal completion is not a valid substitute inside a partite construction: a new edge between vertex parts can become an unwanted edge of a designated transversal induced copy.

It is also tempting to name a PEO and use its two relative orientations on an edge as a Ramsey expansion. The next exact diagram blocks that inference.

Proposition 6.3 (A PEO-amalgamation obstruction). *The class of chordal graphs equipped with a named PEO fails amalgamation under induced embeddings.*

Proof. Let $A = \{p, q\}$ be a nonedge with $p \prec q$. Let B_1 have PEO $p \prec x \prec q$ and edges px, xq . Let B_2 have PEO $y \prec p \prec q$ and the single edge yq . Both extend the same PEO-expanded copy of A .

In any amalgam, the inherited orders force $y \prec p \prec x \prec q$. At q , its earlier neighbors x, y must be adjacent, so xy is forced. At x , its earlier neighbors p, y must then be adjacent, so py is forced. This contradicts the requirement that the copy of B_2 be induced. $\square$

Thus the PEO coloring in Proposition 6.1 is a genuine lower-bound device, but naming a PEO does not by itself give a Ramsey class or an upper bound of two. A successful proof for all chordal graphs would need a partite construction that preserves induced chordality without relying on this false amalgamation premise. That is the unresolved step.

7. REPRODUCIBILITY AND DISCLOSURE

The lower-bound witness in (2) is finite and can be checked by hand. The supplementary material includes a machine-readable copy and an independent exact verifier that enumerates all of its split partitions and recomputes the three ordered edge types. The verifier uses only integer and Boolean operations; the theorem does not rely on numerical experiments.

Computer assistance was used for finite witness checking, bibliographic searches, typesetting, and consistency review. AI assistance was used to help organize these tasks and edit the exposition. All mathematical claims were rederived from the definitions, the cited Ramsey theorem was checked against its stated hypotheses, and the final proof was audited independently at the definition and quantifier level. No proof assistant was used.

LIMITATIONS

Theorem 1.1 concerns split hosts only. It does not determine the small Ramsey degree in the full chordal class. The literature-priority claim is also deliberately bounded: the searches described above found no prior explicit determination of the ordered split-graph value, but cannot exclude unpublished work or work indexed under different terminology.

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INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY, KUALA LUMPUR, 56000, MALAYSIA

Email address: zijianzeng@foxmail.com

Email address: 1002266693@ucsiuniversity.edu.my